

The OpenTelemetry mistakes I keep seeing

And how to stop making them

Juraci Paixão Kröhling · WeAreDevelopers 2026



■ ■ Juraci Paixão Kröhling



OpenTelemetry

Governance Committee member and long-time project contributor.

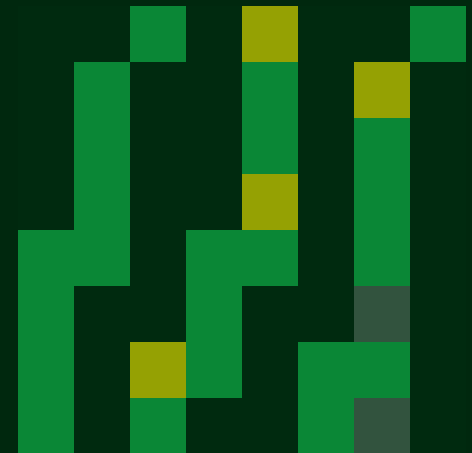


OllyGarden

Founder and CEO. Helping teams build telemetry they can trust.



I'll help you and your agents
have telemetry you can trust.



■ ■ Agenda

■ Telemetry primer

The terms and boundaries we need for the examples.

■ Four mistakes

Real, anonymized failures found in production telemetry.

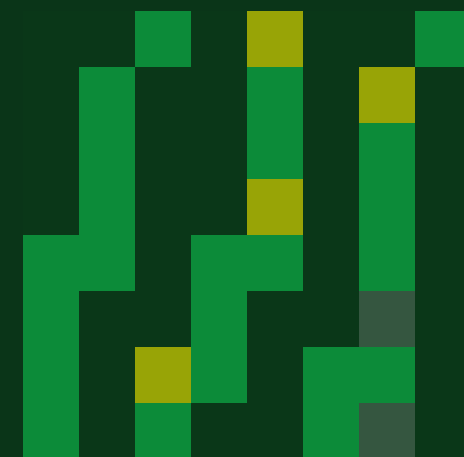
■ Key takeaways

A practical way to find and prevent the same problems.



All examples are real.

Anonymized, but real.



01

Observability primer

Telemetry, monitoring, observability, and
OpenTelemetry



■ ■ Telemetry, monitoring, and observability

Telemetry

The monitoring data you collect from an application.

Monitoring

Dashboards and alerts answer questions you know you will have.

Observability

Queries against telemetry help answer questions you did not predict.

■ ■ OpenTelemetry

Instrumentation

Application code generates logs, metrics, and traces.

Collection

Infrastructure receives telemetry and sends it to a backend.

Standards and protocols

OTLP and semantic conventions keep the parts interoperable.

02

Mistake 1: leaking sensitive data

Telemetry can cross more trust boundaries than application data.



■ ■ Sources of sensitive data leaks

Tools	Admins	Developers	Users
Auto-instrumentation can capture more than the team expects.	A small configuration mistake can disclose secrets to a vendor.	Opaque identifiers are safer than email addresses and names.	User-generated content needs filtering before it becomes telemetry.

Personally identifiable information in telemetry

```
> 2026-07-07 13:42:42.016 INFO Traitement du message reçu avec succès: EvenementSuccesDTO(evento=comunicacao-open-api-exemplo, action=200: Notification envoyée avec succès, messageId=J210119734UYIZ71G8ZON7KS, tipo=EMAIL, broker=EMSG, dataEvento=2026-07-06T00:01:19.875370451Z, identificadores=IdentificadoresSucessoDTO(seletor=000123456789000098), dadosProduto=DadosProdutoDTO(destinatarios=celine.morin@example.com))

: 2026-07-07 13:42:41.922 INFO Traitement du message reçu avec succès: EvenementSuccesDTO(evento=comunicacao-open-api-exemplo, action=200: Notification envoyée avec succès, messageId=J210119725YC3EGJ82LBQ28W, tipo=EMAIL, broker=EMSG, dataEvento=2026-07-06T00:01:19.870068524Z, identificadores=IdentificadoresSucessoDTO(seletor=000123456789000098), dadosProduto=DadosProdutoDTO(destinatarios=chloe.rousseau@example.com))

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```


■ ■ The same leak repeated across log records

```
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■ ■ A search makes the repeated leak visible

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```

Personal data arrives through ordinary application fields



Example courtesy of Elena Kovalenko

■ Signed URLs disclose credentials in attributes

▼ Span attributes

http.request.method	"POST"
http.response.status_code	"202"
network.protocol.version	"1.1"
server.address	"<REDACTED>.e7.environment.api.powerplatform.com"
server.port	"443"
thread.id	"20024"
thread.name	"org.springframework.jms.JmsListenerEndpointContainer#0-4"
url.full	"https://<REDACTED>.e7.environment.api.powerplatform.com:443/powerautomate/automations/direct/workflows/7a1111... sig=***&api-version=1&sv=1.0&sp=%2Ftriggers%2Fmanual%2Frun"

The credential is easy to miss inside a large payload

▼ Span attributes

http.request.method	"POST"
http.response.status_code	"202"
network.protocol.version	"1.1"
server.address	"<REDACTED>.e7.environment.api.powerplatform.com"
server.port	"443"
thread.id	"20024"
thread.name	"org.springframework.jms.JmsListenerEndpointContainer#0-4"
url.full	"https://<REDACTED>.e7.environment.api.powerplatform.com:443/powerautomate/automations/direct/workflows/7a1111...sig=***&api-version=1&sv=1.0&sp=%2Ftriggers%2Fmanual%2Frun"

A JVM trust store password appears in process arguments

▼ Resource attributes

os.description	"Linux 5.14.0-427.87.1.el9_4.x86_64"
os.type	"linux"
process.command_args	["/usr/lib/jvm/java-21-openjdk-21.0.10.0.7-1.el8.x86_64/bin/java", "-Xms768m", "-Xmx768m", "-Djava.security.egd=file:/dev/./urandom", "-Duser.timezone=Europe/Berlin", "-Djavax.net.ssl.trustStorePassword=***", "-Djavax.net.ssl.trustStore=/home/jboss/trusted.j", "org.springframework.boot.loader.launch.JarLaunch
process.executable.path	"/usr/lib/jvm/java-21-openjdk-21.0.10.0.7-1.el8.x86_64/bin/java"
process.pid	"1"
process.runtime.description	"Red Hat, Inc. OpenJDK 64-Bit Server VM 21.0.10+7-LTS"
process.runtime.name	"OpenJDK Runtime Environment"
process.runtime.version	"21.0.10+7-LTS"
	"cd=consigado_cnsq-srv-ins-sme-6c66b7cf7f-66ksh_cns"

■ ■ Process metadata keeps the full command line

```

▼ Resource attributes
os.description      "Linux 5.14.0-427.87.1.el9_4.x86_64"
os.type             "linux"
process.command_args
[
  "/usr/lib/jvm/java-21-openjdk-21.0.10.0.7-1.el8.x86_64/bin/java",
  "-Xms768m",
  "-Xmx768m",
  "-Djava.security.egd=file:/dev/./urandom",
  "-Duser.timezone=Europe/Berlin",
  "-Djavax.net.ssl.trustStorePassword=***",
  "-Djavax.net.ssl.trustStore=/home/jboss/trusted.j",
  "org.springframework.boot.loader.launch.JarLaunch"
]
process.executable.path "/usr/lib/jvm/java-21-openjdk-21.0.10.0.7-1.el8.x86_64/bin/java"
process.pid          "1"
process.runtime.description "Red Hat, Inc. OpenJDK 64-Bit Server VM 21.0.10+7-LTS"
process.runtime.name    "OpenJDK Runtime Environment"
process.runtime.version  "21.0.10+7-LTS"

```

The password is visible to anyone who can query the span

▼ Resource attributes

```

os.description      "Linux 5.14.0-427.87.1.el9_4.x86_64"
os.type            "linux"
process.command_args [
  "/usr/lib/jvm/java-21-openjdk-21.0.10.0.7-1.el8.x86_64/bin/java",
  "-Xms768m",
  "-Xmx768m",
  "-Djava.security.egd=file:/dev/./urandom",
  "-Duser.timezone=Europe/Berlin",
  "-Djavax.net.ssl.trustStorePassword=***",
  "-Djavax.net.ssl.trustStore=/home/jboss/trusted.j",
  "org.springframework.boot.loader.launch.JarLaunch"
]
process.executable.path "/usr/lib/jvm/java-21-openjdk-21.0.10.0.7-1.el8.x86_64/bin/java"
process.pid           "1"
process.runtime.description "Red Hat, Inc. OpenJDK 64-Bit Server VM 21.0.10+7-LTS"
process.runtime.name    "OpenJDK Runtime Environment"
process.runtime.version  "21.0.10+7-LTS"

```


03

Mistake 2: over-instrumentation

More telemetry can reduce the chance of finding the useful signal.



■ ■ Sources of over-instrumentation

Traces

Function-level spans, health checks, and static assets create volume without context.

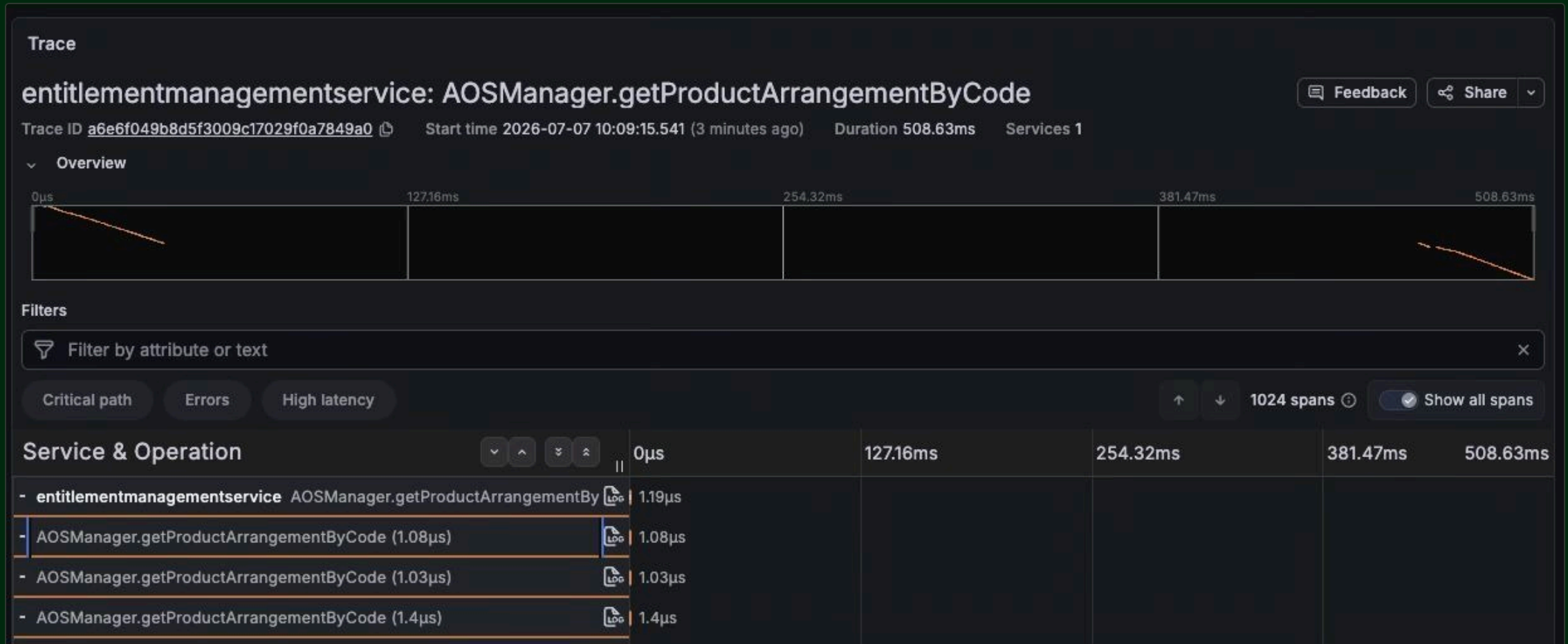
Metrics

Unbounded attributes and aggressive intervals multiply time series.

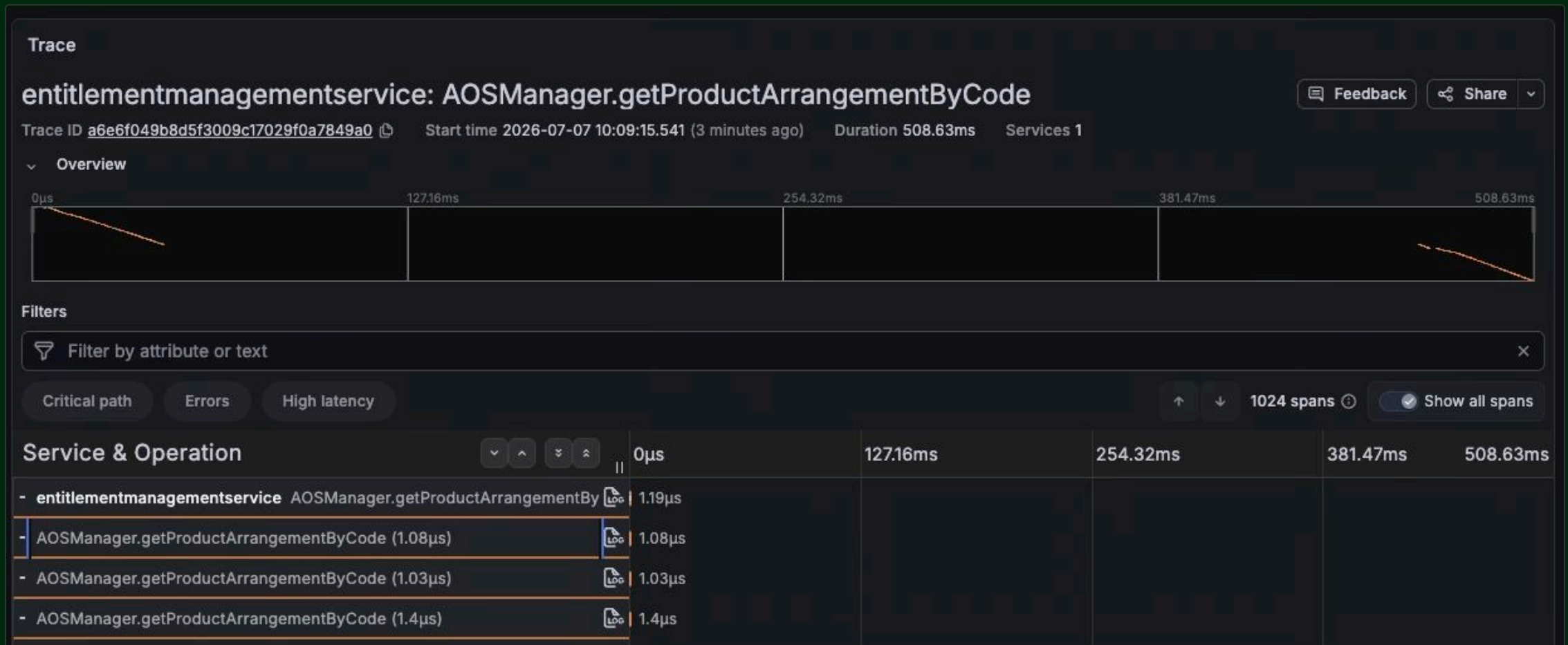
Logs

Debug statements in production hot paths generate repetitive records.

■ ■ One operation can produce a huge trace



■ ■ This example trace contains 1,024 spans



A debug statement in the hot path becomes a volume problem

Log line

```
{  
  "level": 20,  
  "severity": "DEBUG",  
  "time": 1783299600010,  
  "service": "payments-service-api",  
  "context": "***",  
  "msg": "Payout settlement disabled (PAYOUT_SETTLEMENT_ENABLED), skipping scheduled sweep"  
}
```


04

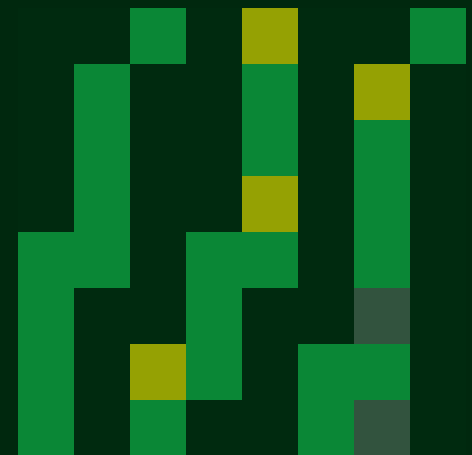
Mistake 3: instrumenting only the happy path

The failure path needs enough context for a tired operator.





What do I need to know at 2 a.m.
when I have not had enough
caffeine?



05

Mistake 4: using the wrong signal

Choose the telemetry type that matches the question.



A counter answers this question better than repeated logs

```
: 2026-07-07 13:12:35.500 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.498 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.498 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.491 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.491 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.491 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.483 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.483 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.482 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.476 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.474 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.474 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.468 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.467 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.466 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.460 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.459 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.459 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
: 2026-07-07 13:12:35.453 WARN Execution of JMS message listener failed, and no ErrorHandler has been set.
```

A trace explains causality better than a list of log lines

```
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```


06

Key takeaways

A short review before telemetry reaches production



Better telemetry starts with deliberate questions

- 01 OpenTelemetry provides the signals needed to understand code.

- 02 Automation handles routine instrumentation, but it also creates noise.

- 03 Some automated output carries sensitive or expensive data.

- 04 Review the telemetry before acting on it.

- 05 Keep each signal only when it answers a useful question.



Thank you

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